



Whole-genome characterization and comparative genomics of a novel freshwater cyanobacteria species: *Pseudanabaena punensis*

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ABSTRACT

Cyanobacteria are emerging as a potential source of novel, beneficial bioactive compounds. However, some cyanobacteria species can harm water quality and public health through the production of toxins. Therefore, surveying the occurrence and generating genomic resources of cyanobacteria producing harmful compounds could help develop the control methods necessary to manage their growth and limit the release contaminants into the water bodies. Here, we describe a novel strain, *Pseudanabaena punensis* isolated from the open ends of pipelines supplying freshwater. This isolate was characterized morphologically, biochemically and by whole-genome sequence analysis. We also provide genomic information for *P. punensis* to help understand and highlight the features unique to this isolate. Morphological and genetic (analysis using 16S rRNA and rbcL genes) data were used to assign this novel strain to phylogenetic and taxonomic groups. The isolate was identified as a filamentous and non-heterocystous cyanobacteria. Based on morphological and 16S rRNA phylogeny, this isolate shares characteristics with the Pseudanabaenaceae family, but remains distinct from well-characterized species suggesting its polyphyletic assemblage. The whole-genome sequence analysis suggests greater genomic and phenotypic plasticity. Genome-wide sequence and comparative genomic analyses, comparing against several closely related species, revealed diverse and important genes associated with synthesizing bioactive compounds, multi-drug resistance pathway, heavy metal resistance, and virulence factors. This isolate also produces several important fatty acids with potential industrial applications. The observations described in this study emphasize both industrial applications and risks associated with the freshwater contamination, and therefore genomic resources provided in this study offer an opportunity for further investigations.

1. Introduction

Cyanobacteria are a unique taxon of Gram-negative oxygenic photosynthetic prokaryotes, which possess diverse morphological forms that range organizationally from unicellular to multicellular (Zehr 2011). Cyanobacteria can grow ubiquitously on Earth, including in extreme environments where light and moisture are present (Whitton and Potts, 2000). Cyanobacteria may be a free-living or live in association with other microbes (such as lichen) and higher taxa (Marchantia, Azolla, Cycads, Gunnera) (Chang et al., 2019). Cyanobacteria exert a

major impact on shaping the environment by fixation of CO₂ and molecular nitrogen and production of oxygen in photosynthesis. Besides, they form the basis for the cycling of nutrients in the aquatic food chain (González-Morales et al., 2020). During the last few decades, cyanobacteria have attracted attention as producers of novel metabolites and have proven importance in industry and medicine. Despite their beneficial roles, because of the instability of the environment and sources of materials, cyanobacteria occasionally become problematic. Under favorable environmental conditions, they can rapidly grow, forming red, blue, or green-colored scums (or cyanobacteria blooms), making it

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difficult for other organisms to survive. Also, they may impart an off-taste and odor to the water and results in the inclusion of several highly toxic compounds in the aquatic system, including sources of freshwater for agriculture and domestic use (Tuji and Niiyama, 2018; Agasild et al., 2019). It has been documented that these toxins cause gastrointestinal disturbances, liver damage, and neural damage in humans and animals (Walker et al., 2019). Some cyanobacteria may thrive in drinking water sources and supplying pipe systems, which in turn may lead to damage substrata from acid production and blocking the filters and pipe systems (Lurling et al., 2015). This highlights the importance of research on harmful cyanobacteria and necessitates knowing the properties and characteristics of those especially growing in freshwater supply systems.

The fossil records of the oldest known cyanobacteria date back approximately 3.5 billion years (Taylor et al., 2009). Currently, cyanobacteria are one of the largest groups of photosynthetic bacteria, and due to the addition of new information, their classification is continually under revision. Earlier, cyanobacteria were classified purely by microscopic observations. However, such morphological observations – made using a microscope – were insufficient to distinguish the cyanobacteria taxa (Desikachary, 1959; Rippka et al., 1979; Shih et al., 2013). Analyses of physiology, cell contents, DNA characteristics, and 16S rRNA genes led to the discovery of new taxonomic groups and allowed reorganization of the classification of cyanobacteria (Komarek and Anagnostidis, 1986; Shih et al. 2013). However, these approaches were also inadequate to resolve the species level relationships (Boyer et al. 2001). Recently, trends have changed, and researchers have begun to use polyphasic approaches to include information from morphology, ecology, biochemistry, and molecular biology. It is established that the 16S-23S rRNA internal transcribed spacer (ITS) is a highly conserved region.

Thus, it has become possible to distinguish and show the phylogenetic relationships between closely related species (Komárek and Anagnostidis, 2005; Martins and Branco 2016).

Furthermore, advances in sequencing technologies have accelerated the identification of the novel organisms and functional variations and facilitated trait discoveries in microorganisms (Caputo et al. 2019), plants (Valliyodan et al. 2016), and humans/animals (Appels et al. 2014). In this study, by applying polyphasic approaches, we present a novel, filamentous, non-heterocystous cyanobacteria species of *Pseudanabena* isolated from the open ends of PVC pipelines supplying freshwater to Pimpri-Chinchwad Municipal Corporation (PCMC), Maharashtra, India.

The specific descriptions are provided following the International Code of Nomenclature for Algae, Fungi, and Plants (Shenzhen Code) (Turland et al., 2018). This study also aims to determine the potential of this cyanobacteria to produce useful bioactive compounds, its multi-drug resistance heavy metal stress tolerance, and toxin production, and offer valuable genomic data for future studies and functional characterizations.

2. Materials and methods

2.1. Sampling site

The isolate was collected from the open ends of polyvinyl chloride (PVC) pipes of supplying freshwater in Charholi, Pimpri-Chinchwad Municipal Corporation (PCMC) area, Pune, Maharashtra, India (Latitude 18.64435625° N; Longitude: 73.92043594° E; collected in the month June–August of 2018–19). Samples were collected from few centimetres inside from the open end of the PVC pipe. Two individual

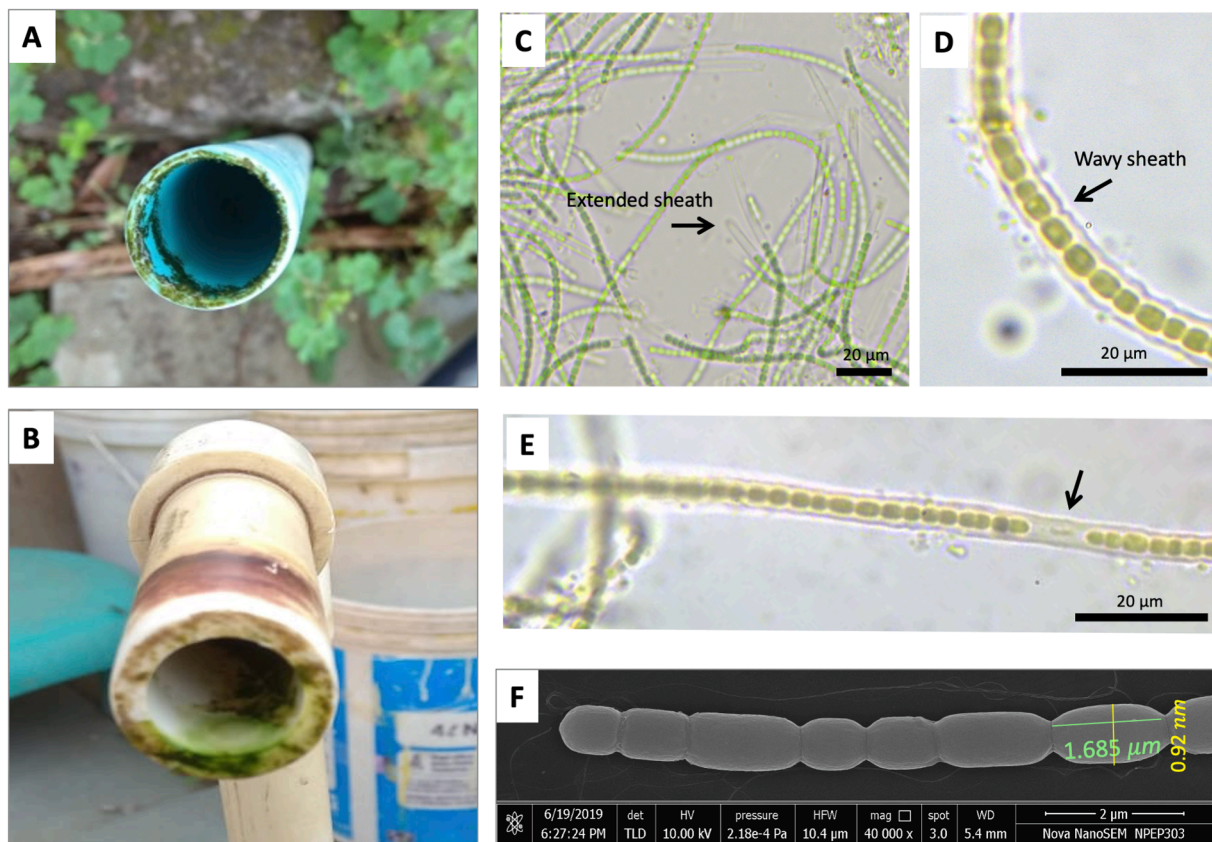


Fig. 1. Collection site of cyanobacteria isolate from open end of PVC (A and B). Morphological features of a new genus *P. punensis* under (C) light microscope showing extended sheath after terminal cell; (D) filament showing wavy sheath; (E) filament showing disintegration of trichomes in hormogonia without necrotic cells (see arrow); (F) scanning electron microscope. The length and width of representative cell is noted.

open ends (Fig. 1 A and B) at a distance of 9.5 m were sampled, which provided the same isolate. The samples were collected from the open end of the pipe using sterile forceps and brought to the laboratory in sterile 5-ml centrifuge tubes.

2.2. Culture of cyanobacteria

For purification and establishment of axenic monoclonal culture, the samples were washed several times by suspending in sterile drinking water, followed by centrifugation at 2500 rpm for 10 min. Individual filaments were inoculated on 1% agarose medium in Petri dishes. Then, from developed colonies, after 1 week of initiation of culture, a single filament was picked and sub-cultured on a fresh medium. The colony development procedure to subculture an individual filament was repeated at a 1-week interval. The growth of filaments was observed from time to time until growth of an axenic monoclonal colony. To establish the isolated culture in a liquid medium, individual filaments from axenic culture from the Petri dish were inoculated into a 300 ml glass bottle containing 100 ml liquid BG-11 medium. The inoculated bottles were placed on a static culture rack. The cultures were sub-cultured in the same fresh medium every 4 weeks and maintained over 8 months. The nutrient formulation recommended by Rippka et al. (1979) was used for the establishment and maintenance of culture. Before the addition of agar-agar and autoclaving at 121 °C for 14 min, the pH of the culture medium was adjusted to 7.5. All the cultures were maintained under controlled growth conditions, i.e., 25 ± 2 °C and 8 h light and 16 h dark cycle (white fluorescent lamp, 30 μmol photons m⁻² s⁻¹).

2.3. Culture deposition

The axenic culture of *Pseudanabaena punensis* sp. nov. was submitted for deposition at National Collection of Industrial Microorganisms, Pune, India (<https://www.ncl-india.org/files/NCIM/>). The strain replicates were also preserved in formaldehyde (4%) and live-stock culture in glycerol at -20 °C in laboratory at the Department of Botany, Hutatma Rajguru Mahavidyalaya, and at Department of Botany, Savitribai Phule Pune University, Pune, MS, India.

2.4. Morphological and ultrastructural evaluation

The sample was brought to the laboratory and established axenic sample cultures were observed under a stereo zoom microscope (Olympus, Japan) as wet mounts. Photomicrographs were taken using a digital camera (UCMOS14000KPA, operated by ToupView Software). A field emission-scanning electron microscope (Fe-Sem, Fei Nova Nano-SEMnp303) (Software: xT microscope) operating at 5–10 kV, 3.5 spot size, and ETD detector were used for observation. For SEM examination, the specimen was fixed with 1.25% glutaraldehyde for 5 h and washed with phosphate buffer saline solution. Then, fixed samples were washed with phosphate buffer saline three times and subsequently dehydrated in graded series of ethanol (20%, 50%, 75%, and 100%). The dehydrated samples were transferred into HMDS (Hexamethyldisilazane) and ethanol (1:2) solution for 2–3 min. the 1:2 solutions were replaced with a 2:1 solution of HMDS and 100% ethanol. The sample was allowed to dry for 2–3 h. The isolate samples were observed for the type of filaments, cellular length, and their width. These morphological characters were considered for taxonomic identification.

2.5. DNA extraction and PCR amplification

Genomic DNA was isolated using Qiagen DNeasy kit. Bacterial 16S region gene was amplified using standard PCR reaction. The primer pair CYA106F and CYS781R was used in a PCR reaction with an annealing temperature of 60 °C. After amplification, products were purified by using Invitrogen PCR product purification kit (Life technologies, USA)

and were directly sequenced using an ABI PRISM BigDye Terminator V3.1 kit (Applied Biosystems, USA). The sequences were analysed using Sequencing Analysis 5.2 software. BLAST analysis was performed at BlastN site at NCBI server (<http://www.ncbi.nlm.nih.gov/BLAST>).

2.6. Phylogenetic analysis

Phylogenetic analysis for the obtained sequences was conducted using the MEGA 7 computer program (Kumar et al., 2016). The alignment was performed using the ClustalW multiple sequence alignment program available at <https://www.genome.jp/tools-bin/clustalw>. Alignments were checked manually. All positions containing gaps and missing data were partially deleted (site coverage cutoff 60%). A Neighbor-Joining tree was calculated using MEGA software with the best fit model determined by Bayesian information criterion scores, and the software calculated the substitution nucleotide matrix parameters.

2.7. Pre-processing of raw reads, genome assembly, and annotation

Whole-genome sequencing was performed using the Illumina HiSeq platform. After sequencing, the quality of paired-end raw reads was checked using the FASTQC tool (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Adapter contamination (if present) and low-quality reads (Q < 30) were trimmed out using NGSQC Toolkit v2.3.3 (Patel and Jain, 2012). If 80% of the read had the average good quality Q ≥ 30, it was kept and called high quality (HQ) data. Genome assembly was done using the SPAdes-3.11.1 assembler with the default option (Bankevich et al., 2012). Genome assembly quality was checked using QUAST (Gurevich et al. 2013), and several rRNAs were identified and fetched out using rnammer-1.2 (Lagesen et al., 2007).

After genome assembly, genome annotation of the organism was done using PROKKA (Seemann, 2014). The tRNAs, rRNA genes, and protein-encoding genes of the genomes were predicted using tRNAScan-SE (Schattner et al. 2005), RNAmmer (Lagesen et al., 2007), and Interproscan (Jones et al., 2014), respectively. The genome's signal peptide prediction was identified using a SignalP finder (Almagro Armenteros et al., 2019). The presence of secondary metabolites was identified using AntiSMASH (Blin et al., 2017). The predicted 16S rRNA sequence was used in the EzBioCloud server (Yoon et al., 2017) to find closely related organisms. Phylogenetic analysis was constructed using MEGA (Kumar et al., 2016). BUSCO (conda version 4.1.4) (Simão et al., 2015) was used to assess the gene space completeness of assemblies using the cyanobacteria_odb10 (2020-03-06), synechococcales_odb10 (2020-03-06), and nostocales_odb10 (2020-03-06) datasets.

2.8. Comparative genome analysis

To compare our complement of genes predicted from our assembly against protein predictions in other cyanobacteria, protein sequences from strains were obtained from Integrated Microbial Genomes at DOE Joint Genome Institute (www.jgi.doe.gov) and NCBI (www.ncbi.nlm.nih.gov/genome). BLASTP searches were performed against the proteins with an e-value of 1E-4, and the top hit for each protein was used to further assess the uniqueness of the new genome assembled in this study. BUSCO scores of *P. punensis* were compared to those of a Synechococcales *Halomicronema hongdechloris* C2206 (GCA_002075285), and a nostocales, *Nostoc azollae* (*Trichormus azollae*) 0708 (GCA_000196515), whose genomes were downloaded from EnsemblBacteria (release 49) (Howe et al., 2020). Synteny analysis was performed using Synima (Farrer 2017) with the OrthoMCL default options.

2.9. Antibiotic-resistant and virulence factor prediction

The presence of antibiotic resistance genes was determined by CARD (Comprehensive Antibiotic Resistance Database). A BLAST search was performed against VFDB (Virulence factor database) with an e-value cut-

off of 0.0001, bit-score ≥ 50 , and percent identity ≥ 50 .

2.10. Sequence submission

The assembled sequences (>200 bp) obtained in the study was deposited in NCBI under the following BioProject ID: PRJNA655233.

2.11. Fatty acids extraction and profiling using GCMS

The method described by Zhu et al. (2002) was used for the extraction of fatty acids. At the end of the fourth week of culture, the biomass of the isolate grown in BG-11 medium and drinking water was harvested by centrifugation (8500 g for 5 min) and washed with sterile distilled water. The biomass was freeze-dried and suspended (0.5 g) in a 20 ml mixture of chloroform and methanol (1:1 v/v). Then, the sample mixture was incubated in a water bath for 30 min at 60 °C, followed by sonication (Digital Ultrasonic Cleaner, CD-4820) at the same temperature for 1 h. The resulting extracts were filtered through PTFE filters (0.2- μ m) and evaporated to dryness using a rotary evaporator under vacuum at 60 °C. The dried extracts were dissolved in 1 ml of acetone and 0.5 ml of 3 M methanolic HCl and then heated at 60 °C for 1 h in a thermostat water bath for derivatization. Next, the samples were cooled at room temperature, followed by the addition of 1 ml of water to obtain phase separation and extracted using 1 $\times$ 3 ml of hexane. Extracts were mixed and dried over anhydrous sodium sulfate, followed by solvent evaporation to dryness. Then, the resulting extracts – Fatty acids methyl esters (FAME) – were dissolved in 1 L of HPLC grade hexane, filtered through a PTFE filter (0.2- μ m), and used for GCMS analysis. After harvesting of biomass of isolate, the filtrate remains of BG-11 medium, and drinking water medium was also lyophilized, extracted, and analyzed by following a similar procedure for exudation of metabolites.

The fatty acid methyl esters were analyzed by GC–MS (Shimadzu, QP 2010 series, Tokyo, Japan) equipped with model AOC-20i autosampler and RTX-1 fused silica capillary column (30 m length, 0.25 mm id, and 0.25 μ m thickness). Three microliters of the sample were injected in GC through a 10- μ l syringe in split mode. It was maintained at 250 °C. The ultrapure helium gas was passed through the GC column with a 2.0 ml/min flow rate. The temperature of the GC oven was set at 50–300 °C with an increment rate of 10 °C/min and hold times of 2 min and 8 min at the initial and final temperature, respectively. The temperature of the thermal Aux 2 (MSD Transfer Line) was set at 250 °C. The mass spectra were recorded at 1.02 kV. The generated amplified signals were saved in a computer using the GC-solution software. The chemical components were identified by comparing their mass fragmentation pattern to NIST libraries' standard reference data.

2.12. Phycobiliprotein

Phycobiliproteins were extracted by suspending the wet biomass (0.1 g) in 0.1 M potassium phosphate buffer (10 ml, pH 7.2), followed by freezing (–20 °C for 1 h) and thawing (28 °C) three times, then sonication (Digital Ultrasonic Cleaner, CD-4820) for 5 min. The resulting slurry was placed in the dark for 48 h, and then after centrifugation (10,000g, 10 min), the obtained supernatant was used to measure the absorbance (Shimadzu UV–VIS spectrophotometer). The determination of phycobiliproteins-phycocyanin (C-PC), allophycocyanin (C-APC), and phycoerythrin (C-PE) were carried out using the equation given by Bennett and Bogorad (1973) and the extinction coefficient from Bryant et al. (1979).

3. Results and discussion

3.1. Morphological characterization and classification

The cyanobacteria isolate was collected from the open ends of a PVC pipe supplying freshwater (Charholi, PCMC area, Pune, India; Fig. 1).

The Pavana Dam is the major source of freshwater supply to this area (Mane et al., 2013), suggesting as a natural freshwater habitat to this isolate. However, analysis of water samples from Pavana river, a municipal water purification center, and local reservoirs and monitoring seasonal water samples is required to understand its dominance and prevalence in drinking water. In this study, samples were collected from two sites (at a distance of 9.5 m) as a single strain that showed characteristic features of the family Pseudanabaenaceae (Table 1 and Table S1). However, compared to the morphological characteristics, the isolate did not resemble any earlier described species present in Algaebase and the National Collection of Industrial Microorganism (Table S1; Castenholz et al., 2015). The phylogenetic investigation indicates that this isolate forms a distinct ancestry within the Pseudanabaenaceae family, considering the nucleotide sequences of the 16S ribosomal RNA (16S rRNA) gene. The phylogenetic analysis and results are discussed in the next section of the manuscript. Based on morphological observations and phylogenetic studies, a new non-heterocystous filamentous *Pseudanabaena* species from the family Pseudanabaenaceae has been illustrated. Therefore, henceforth, the isolate is referred to by the name *Pseudanabaena punensis*, sp. nov. and abbreviated as *P. punensis*.

3.2. Taxonomic descriptions

Class: Cyanophyceae

Order: Synechococcales

Family: Pseudanabaenaceae

Genus: *Pseudanabaena*

Species: *punensis* Nitnaware et Nikam sp. nov.

Habitat: Freshwater

Type locality: Charholi BK, Pune, India

Collection site: Open ends of PVC pipelines

Geographical coordinates: Latitude 18.64435625° N

Longitude: 73.92043594° E.

Holotype: The isolate *Pseudanabaena punensis* Nitnaware et Nikam sp. nov. was deposited at the National Collection of Industrial Microorganisms, Pune, India (<https://www.nci-india.org/files/NCIM/>).

Etymology: The specific epithet reflects the occurrence of the species in the Pune district in MS, India.

3.3. Morphological observations

The filaments in liquid culture adhere to the culture bottle wall and form a light green macroscopic aggregate. Using a single filament from 6-week-old cultures, on transfer to the surface of the solid medium, we observed growth in the form of a cluster of filaments which gradually formed bright blue-green colonies. Gliding motility was observed in filaments on the smooth surface of a solid medium. Under the light microscope, from both solid and liquid cultures, filaments appeared straight and long, without any branching, loosely entangled, and sparsely lonely (Fig. 1C–E). Cells in the filaments were cylindrical, thin, colorless, and mucilaginous wavy sheath that extends beyond the terminal cell of the trichome (Fig. 1D). The cell length and width of *P. punensis* showed similar pattern with *P. frigida*, however, other morphological features and genomic content (estimated genome size and gene number) were largely distinct (Table S1). Cells within the trichome were longer than wide, deeply constricted at the cross wall, and a gas vesicle was observed. Terminal cells were also elongated and slightly pointed at the apex (Fig. 1F). Morphological traits and dimensions of cells were compared with diverse and closely related accessions (Table 1 and Table S1, respectively). Using a scanning electron microscope, we observed that cell length ranged from 0.77 to 2.3 μ m and cell width ranged from 0.82 to 1.2 μ m. The terminal cell size was 0.86 – 1.6 (L) and 0.77–1.06 (W).

Phycobiliproteins [phycoerythrin (PE), phycocyanin (PC) and allophycocyanin (APC)] are light energy capturing proteins and are used to identify and differentiate the cyanobacteria from other planktonic

Table 1

Comparative ecological and morphological characters of some of the selected cyanobacterial species. Abbreviations- W: width, L: length, IC: Involution Cell, NS: Not stated.

#	Organism	Trichome	Habitat	W μm	L μm	Apical cells	Constricted walls	Motility	Sheath	Gas vacuole	Hormogonia	IC	Ref
1.	<i>Limnothrix</i>	Straight, curved, cells cylindrical or barrel shaped	Mesotrophic to eutrophic freshwater reservoirs	1–6	3–10	Cylindrical (round apex)	+	+	–	+	(obligate)	NS	Azevedo et al 2003
2.	<i>Pseudanabaena</i>	straight slightly curved, solitary	Aquatic and benthic Brackish and fresh water, terrestrial	1–6	2.1–1.9	Rounded (conical apex)	+/-	+	–	+/-	NS	NS	Zimba et al. 2016
3.	<i>Amazoninema</i>	Filaments entangled, flexuous	Oligotrophic freshwater	1.2–2.2	1.1–3.0	Cylindrical (round apex)	+	+	+	–	(Rare)	+	Genuário et al 2018
4.	<i>Toxifilum</i>	Straight or coiled	Marine	1.3–3.3		Rounded	+	NS		–	+		Zimba et al 2016
5.	<i>Pseudanabaena biceps</i> , PCC 7429	Straight and longer	Freshwater, Wetlands	2.5	–	Round and pointed	+	+	+	-/+	+	NS	Shih et al. 2013
6.	<i>Pseudanabaena foetida</i>	Straight and longer	Freshwater	1.0–1.5	3.9–11.0	Rounded	+	+	+	–	+	NS	Niiyama et al. 2016
7.	<i>Pseudanabaena amphigranulata</i>	Straight and cylindrical	Freshwater, man-made ponds	1.2–2	2–5	Rounded	+	–	–	–	+	NS	Khan et al. 2018
8.	<i>Pseudanabaenaceae cyanobacterium CENA510</i>	Straight and cylindrical	Marine	NS	NS	NS	NS	NS	NS	NS	NS	NS	Andreote et al. 2014 ; Joint Genome Institute (jgi.doe.gov)
9.	<i>Leptolyngbya</i> sp. PCC 7376	NS	Aquatic and fresh water, terrestrial	NS	NS	NS	NS	NS	NS	NS	NS	NS	Shih et al. 2013 ; Joint Genome Institute (jgi.doe.gov)
10.	<i>Pseudanabaena</i> sp. ABRG5-3	Semi-filamentous	Fresh water, pond	–	–	ellipse or round shape	+	+	–	–	+	NS	Kitazaki et al. 2013
11.	<i>P. punensis</i>	Straight and longer	Drinking water PVC pipeline	0.88	1.38	Cylindrical (pointed apex)	+	(deep)	+	+	–	+	Present Study

species (Vincent et al. 2004; Everroad and Wood, 2006; Zimba et al. 2016; Basheva et al. 2018). The analysis of absorption spectra in *P. punensis* identified phycocyanin (PC; 11 ± 0.02 mg/ml), allophycocyanin (APC 6 ± 0.01 mg/ml) and phycoerythrin (PE; 2 ± 0.03 mg/ml). *P. punensis* has elevated PC concentrations relative to the other phycobilins. Using the AlgaeBase database, we compared *P. punensis* with 36 *Pseudanabaena* species (Table S1). The morphological features of the isolate (Table S1) suggest that some morphological features of the newly identified *P. punensis*, such as straight filament, non-heterocystous, cylindrical cells, resemble other species in the family Pseudanabaenaceae. However, they largely differ in cell dimensions, terminal cell structure, motility, and extended and wavy sheath structure. Heterocyst was not observed along the filaments. Terminal cell morphology is one of the principal criteria for traditional classification (Komárek and Anagnostidis, 2005), and in this study, we identified a slightly pointed terminal, while some cells were appeared round compared to typical conical shape in other *Pseudanabaena* species (Fig. 1). We used molecular and genomics approaches to characterize this new species to elaborate and support our morphological findings.

3.4. Phylogenetic analysis of 16S rRNA and *rbcL* genes

The 16S rRNA is ubiquitous in nature and extensively used as a standard genetic marker for classification, evolutionary studies, and identification of Bacteria and Archaea (Kim and Chun 2014). Previously,

Komárek and Anagnostidis (2005) and Andreote et al. (2014) reported the classification of new species based on 99.7% identity between different species of *Pseudanabaenales*. Similarly, species of *Phormidium* sp. were classified using 16S rRNA having 99% identity (Andreote et al., 2014). To better understand the sequence diversity and evolutionary relationship between different cyanobacteria species, we sequenced the 16S rRNA from *P. punensis* and constructed a phylogenetic tree by including 35 16S-rRNA representing 24 different cyanobacteria species (Fig. 2). In these methods, the reliability of trees was assessed by performing 1000 bootstrap replicates (Tamura et al., 2011). The sequences used in this study represent a broad range of morphologies, habitats, and metabolisms. In addition to phylogenetically and phenotypically diverse cyanobacteria strains, we included two 16S rRNA sequences from closely related *Pseudanabaenaceae* strains, including *Pseudanabaenaceae* PCC7367 and *Pseudanabaenaceae* CENA510 (Andreote et al. 2014). The phylogenetic analysis showed two distinct clades (Fig. 2), where the newly identified genus *P. punensis* clustered with the *Synechococcales* order, namely *Pseudanabaena* CENA510 (98% identity) and *Halomicronema hongdechloris* (92% identity), supporting the morphological characters associated with *Synechococcales*. Interestingly, *Pseudanabaena* CENA510 was collected from extreme saline-alkaline environmental conditions (Andreote et al. 2014), while *P. punensis* was collected from fresh water. Clade I represented most accessions from filamentous cyanobacteria, and clade II represented accessions from all three classes, including filamentous, unicellular, and heterocystous cyanobacteria. In

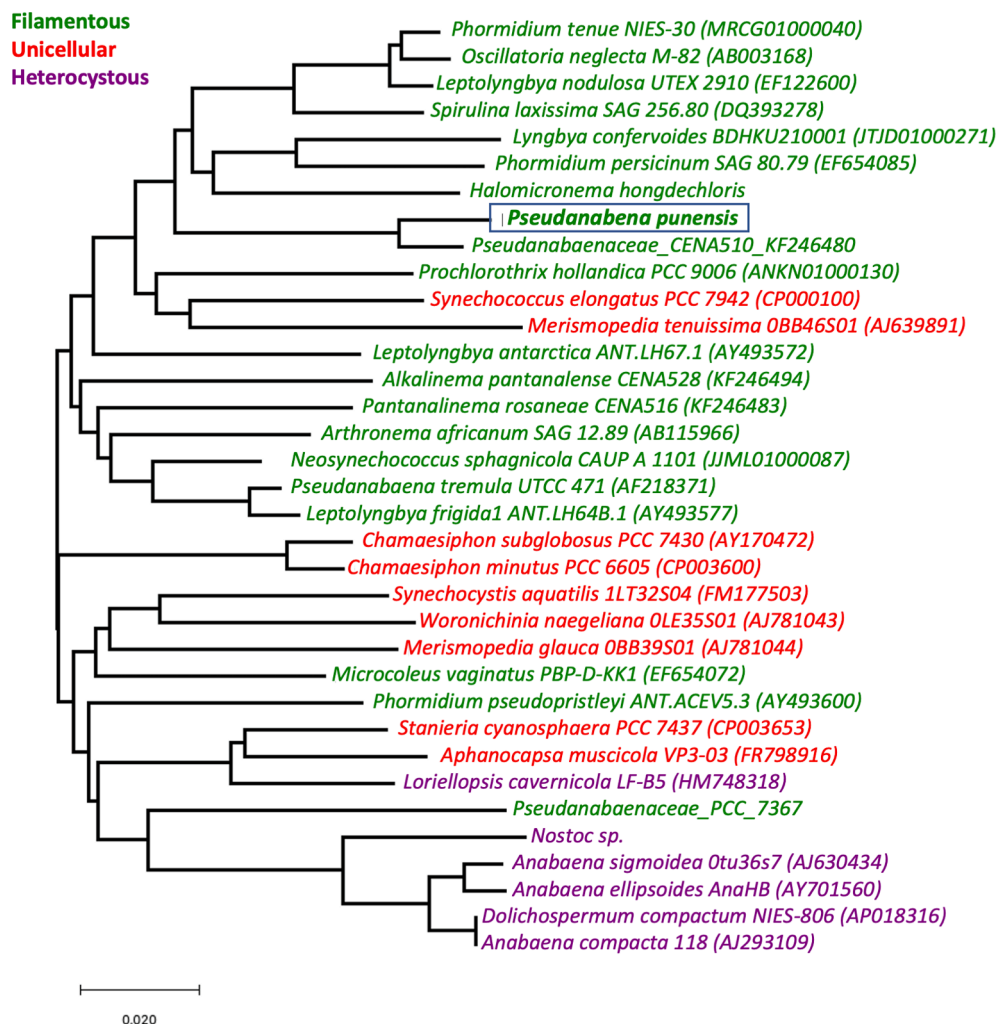


Fig. 2. Maximum-likelihood phylogenetic analysis of 16S rRNA in selected cyanobacteria species. Branches are color coded according to morphological subsections as filamentous, unicellular and heterocystous. Taxa highlighted with box was sequenced in this study.

agreement with Andreote et al. (2014) and Shih et al. (2013), this observation suggests the functional plasticity associated with evolutionary conservation of Synechococcales/Pseudanabaenaceae members further facilitated the adaptation and diversification to extreme environments. Furthermore, a complementary phylogenetic tree was constructed using a large subunit of ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCo) *rbcL* genes from over 100 accessions (Supplementary Figure S1). The *rbcL* gene is the key and conserved gene involved in the first major step of carbon fixation in plants and is commonly used in phylogenetic analysis and evolution studies (Elsaied and Naganuma, 2001; Newmaster et al. 2006). The major subclades of derived by *rbcL* gene phylogenetic analysis were highly congruent with the 16S rRNA phylogenetic tree. The *P. punensis* *rbcL* gene was clustered with other *Pseudanabaena* sp. and *Leptolyngbya* sp. (Supplementary Figure S1). It has been reported that pseudanabaenacean strains have morphological features similar to the genus *Leptolyngbya* sp., consisting of strains with thin trichomes and isodiametric cells. *Leptolyngbya* sp. are considered as a polyphyletic assemblage (Andreote et al. 2014). These facts, along with phylogenetic relatedness with *Pseudanabaenaceae* and *Leptolyngbya* sp., suggest that *P. punensis* could also be polyphyletic; however, more data on related strains are needed to infer these observations.

3.5. Genome sequencing, assembly, and annotation

The morphological observations and phylogenetic analysis using 16S rRNA and *rbcL* substantiated new species isolated from PVC pipes (supplying freshwater). Whole-genome sequencing (WGS) offers several advantages in examining the genomic complexity of newly identified species and comparing genomes across multiple species or samples to identify novel genes and allelic variants (Patil et al., 2019). Therefore, to better understand the genome complexity and genomic context of *P. punensis*, we performed WGS and compared its genomic features with other closely related cyanobacteria genomes. Sequencing of *P. punensis* DNA yielded a total of 10.02 million 151-bp length paired-end reads and a total of 1.51 billion bases with approximately $220 \times$ genomic coverage. Following the quality testing of raw reads, clean sequence reads were assembled *de novo* using the St. Petersburg genome assembler (SPAdes-3.11.1; Bankevich et al. 2012). The quality assessment of assembled genome was performed using QUAST tools (Gurevich et al., 2013; Supplementary Figure S2), and rRNA identification was performed using RNAmmer-1.2 tools (Lagesen et al. 2007). A total of 49 contigs with a maximum length of 0.99 Mb and 24 contigs with more than 5 Mb were identified (Supplementary Figure S2). The assembly of contigs had $N_{50} = 4,50,637$. The estimated genome size was 6.85 Mb (6,844,194 bp), with an estimated G + C content of 56.71% (Table 2). The estimated genome size of cyanobacteria ranges from 1.67 Mb (*Prochlorococcus marinus*) to 12.3 Mb (*Scytonema hofmanni* PCC 7110), while the estimated genome size within *Pseudanabaena* ranges from 4.8 to 5.6 Mb (Shih et al. 2013). This observation suggests that *Pseudanabaenaceae* not only cover a broad range of habitat and morphology but the genomic content also varies greatly within the same order.

Table 2
Assembly and Contigs statistics of *P. punensis* genome.

Attributes	Value
Genome size (bp)	68,56,169 (6.85 Mb)
G + C content	56.71 %
N50 length (contigs)	3,66,997
Total annotated genes	6,056
Protein encoding genes	5,966
Protein-encoding genes with predicted functions	5,713
Total number of predicted misc-RNA	38 (compare)
Total tRNAs	48 (compare)
16S rRNA	1
30S RNA	26

Therefore, sequencing diverse cyanobacterial species would be essential to uncover the genomic features and discovering novel traits.

The assembled genome was annotated using two independent annotation tools, PROKKA (Seemann, 2014) and Omics Box (<https://www.biobam.com/omicsbox/>). We identified 6056 genes, with 5966 genes encoding for protein and about 5713 protein-encoding genes with predicted function. The average protein length was estimated to be 325 amino acids. Notable highly represented supergene families are those for P-loop containing nucleoside triphosphate (357 genes), CheY-like superfamily (161 genes), and HSP90-ATPase (136 genes) were identified. Furthermore, cell constituent (CC) categories were enriched for cellular component (456 genes), cytoplasm (129), and membrane transport (118 genes); whereas those for biological processes (BP) were enriched for the biosynthetic process (286 genes); cellular nitrogen compound (204) and small molecular, metabolic process (135 genes); and for molecular functions (MF) genes were enriched for ion binding (412) and molecular function (222) (Supplementary Figure S3). Prediction of non-coding RNA resulted in identifying 48 tRNAs, three rRNAs, one tmRNA, and 38 misc_RNAs. The secreted protein prediction was performed using SignalP 5.0, and we identified 273 secretory proteins (SignalP-noTM 250 and SignalP-TM 23), accounting for 4.57% of total predicted protein (Supplementary Table S2).

3.6. Comparative genome analysis

Based on morphological analysis under light and scanning electron microscopy (Fig. 1) and 16S rRNA (Fig. 2) and *rbcL* gene phylogenetic analysis (Supplementary Figure S1), the newly identified cyanobacteria strain collected from the freshwater supplying pipelines exhibited typical morphological features and sequence similarity to *Pseudanabaena* family under Synechococcales orders. For comparative genome analysis, the predicted protein sequence of *P. punensis* was compared against NCBI non-redundant protein sequence (nr) database, and a total of 2,87,7780 (average 486) hits were obtained using 5996 predicted protein sequences. Among 5966 query sequences, 99% (5,920) showed hits against known 370 organisms listed in taxdb (NCBI taxonomy database). Notably, the top five hits were against Synechococcales (*H. hongdechloris*; *S. elongates*; *Thermoleptolyngbya* sp.; and *L. boryana*) and with one Oscillatoriales (*Geitlerinema* sp.) (Fig. 3A). We also compared the predicted protein sequences of *P. punensis* against protein sequences from the *Pseudanabaenaceae* and *Leptolyngbyaceae* family, which has whole-genome sequence data available. These genomes represent, *P. biceps* PCC_7429 (5.58 Mbp), *P. yagii* (5.57 Mbp), *P. frigida* (5.39 Mbp), *Pseudanabaena* sp. ABRG5-3 (4.7 Mbp), *Pseudanabaena* sp. PCC_7367 (4.89 Mbp), *C. aponium* PCC_20605 (4.11 Mbp), and *Leptolyngbya* sp. PCC_7376 (5.1 Mbp) (Fig. 3B). Similarity analysis using BLASTP among these strains indicated most of the predicted proteins had a poor match, and we found that less than 500 proteins had protein similarity $\geq 80\%$ (Fig. 3B). Nevertheless, *P. punensis* genome resembles closely with three *Pseudanabaena* sp. and one *Leptolyngbya* sp., while *C. Aponium* was distinctly related (Fig. 3B). This analysis suggests that *P. punensis* could be polyphyletic, showing close resemblance with *Pseudanabaenaceae* and *Leptolyngbyaceae* family. However, more detailed research is needed to confirm this inference. Altogether, these results conclude that the species identified in this study belong to order Synechococcales and family *Pseudanabaenaceae* but have no match with any of the previously characterized strains within this family. Furthermore, to obtain a holistic overview of whole-genome diversity, we explore the Microbial Genomes Atlas (MiGA) that offered a robust taxonomic classification of a query genome (Rodriguez-R et al., 2018). In agreement with the NCBI BlastN analysis, MiGA analysis suggested that the *P. punensis* showed close homology with *Halomicronema hongdechloris* C2206 (NZ_CP021983), with 58.69% amino acid identity and *Nostoc azollae* (*Trichormus azollae*) 0708 (GCA_000196515), with a 58.98% amino acid identity. Both *H. hongdechloris* and *P. punensis* belong to the Synechococcales order and share several morphological features

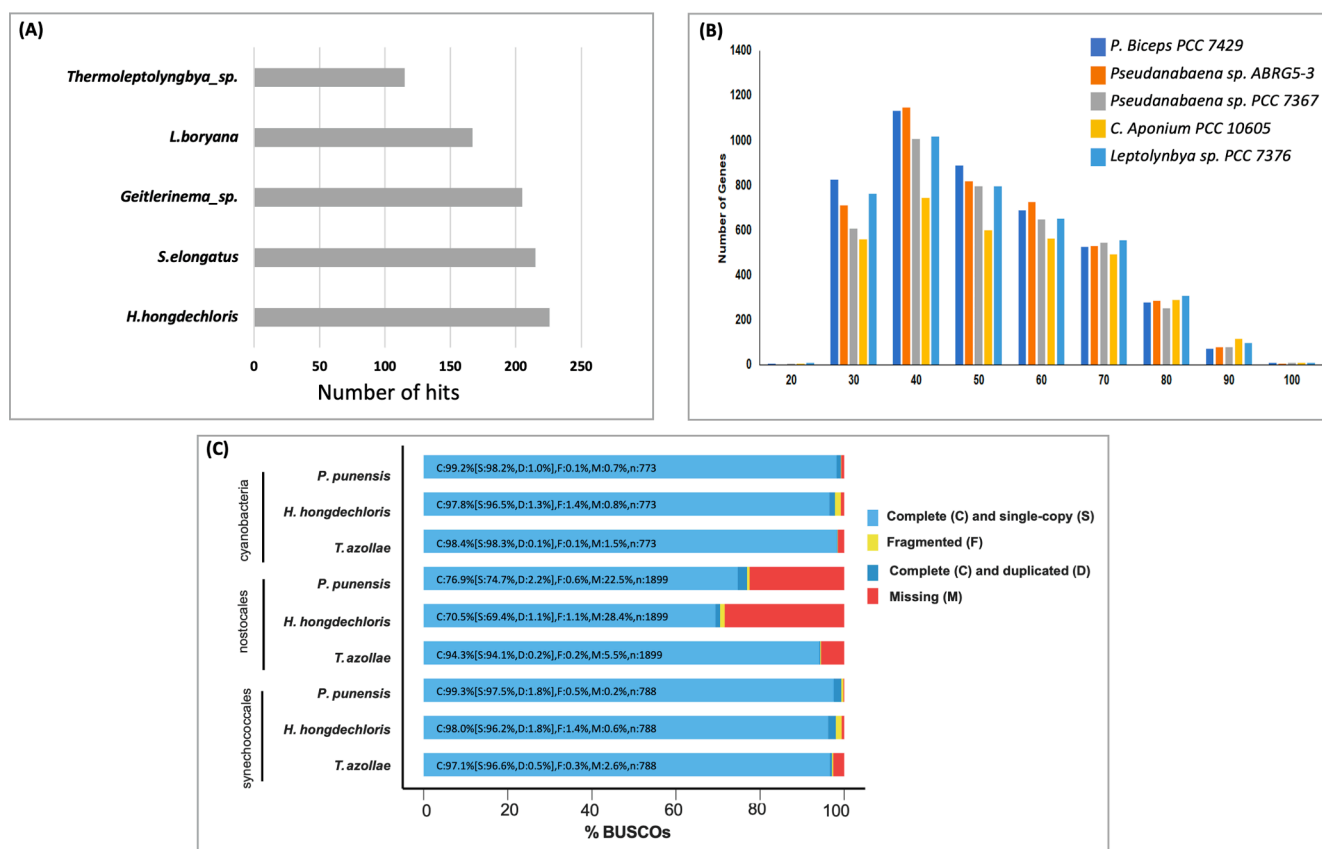


Fig. 3. Comparative genome analysis of *P. punensis* (A) Top five hits after BLASTP of *P. punensis* against NCBI non-redundant database; (B) Comparison of predicted protein sequences with five *Pseudanabaenaceae* accessions having whole genome sequence data; (C) Analysis of conserved single-copy orthologues using BUSCO tool.

(<https://www.algaebase.org/>). BUSCO (Benchmarking Universal Single-Copy Orthologs) analysis of conserved single-copy orthologues is widely used as a proxy for genome completeness and accuracy (Simão et al., 2015). To provide a quantitative measure of the completeness of

genome assembly and annotated gene sets, we performed a BUSCO analysis and identified that the *P. punensis* assembly is of high gene space completeness with BUSCO completeness values of 99.3% against the Synechococcales dataset, while it showed only 76.9% similarity against

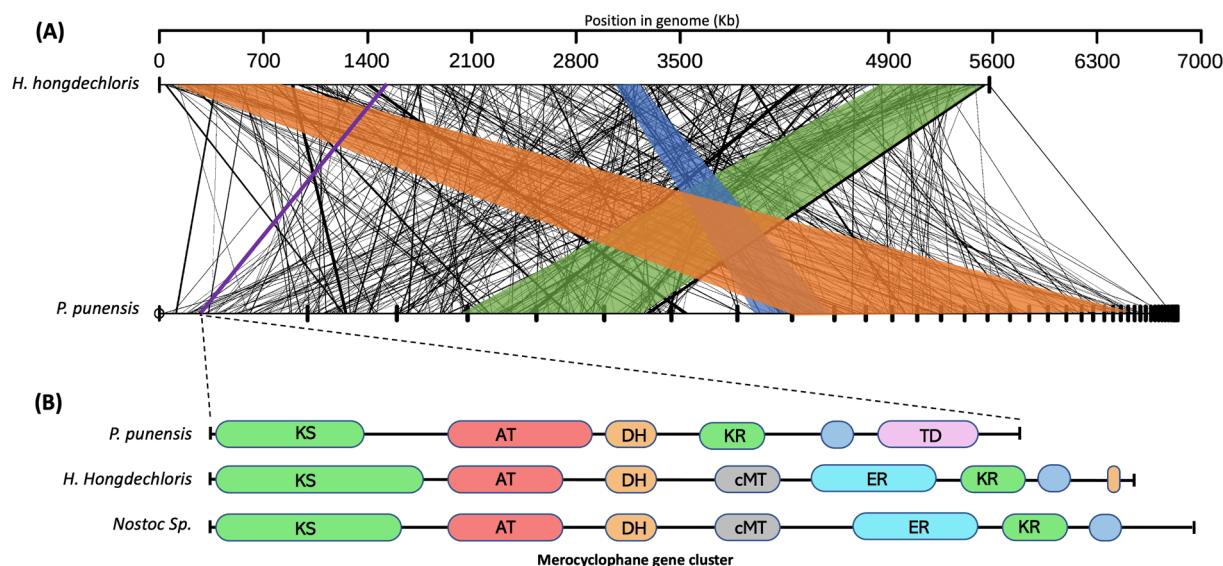


Fig. 4. *P. punensis* genome synteny analysis. (A) Genome-wide synteny analysis between *P. punensis* and *H. hongdechloris*. Microsynteny blocks with collinear gene order are highlighted with orange, blue and green colors. Mercocyclophane gene cluster synteny highlighted with solid purple color line; (B) Comparison of mercocyclophane biosynthetic gene clusters showing similarities in the functional domains and structure of the gene for *P. punensis*, *H. hongdechloris* and *Nostoc* Sp. UIC10110. Abbreviations: KS: Ketide synthase; AT: acyltransferase; DH: dehydratase; KR: beta-ketoacyl reductase; ER: Enoyl reductase; TD: thioester reductase; cMT: methyltransferase. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

nostocales dataset (Fig. 3C). Altogether this analysis substantiated that *P. punensis* is a novel genome and belongs to Synechococcales orders (Fig. 3C). To gain insights on genome conservation, we investigated genome-wide synteny between *P. punensis* and *H. hongdechloris* and reviewed the evolutionary conservation of homologs genes and gene order (Fig. 4). While several genomic blocks or gene clusters are shared between both species (microsynteny), we observed considerable variation in both genomes (Fig. 4A). The syntenic position of the merocyclophane gene cluster was highlighted, and the conserved protein domains are shown in Fig. 4B (details for merocyclophane are provided in the next section).

3.7. Identification of genes involved in secondary bioactive compound pathways

Blue-green algae are important producers of bioactive secondary metabolites (Harada 2004; Singh et al., 2015). To identify genes involved in secondary metabolite production in *P. punensis*, we used the antiSMASH 5.0 tool (Table 3). This analysis identified three potential clusters in the synthesis of Pneumococcal choline-binding protein A (PcpA), Merocyclophane C/D, and 1-heptadecene were significant based on protein domain similarity. The PcpA is a family of surface proteins non-covalently bound to the pneumococcus bacteria (Maestro and Sanz, 2016). However, their role in cyanobacteria remains elusive and requires further investigation. The second bioactive secondary metabolite pathway protein identified was Merocyclophane (*P. punensis* 00613). This protein carries a polyketide synthase (PKS) multi-domain, implicated in bioactive metabolite pathway in plants and other organisms. Over 70% of cyanobacteria encode for PKS gene clusters (Shih et al., 2013). However, few cyanobacteria comprise all associated domains (a cluster responsible for producing merocyclophanes) needed for bioactivity. The protein identified from cyanobacteria (*Nostoc* sp UIC10110 and UIC10062) carries major associated domains and shows antiproliferative activity against cancer cell lines (Kang et al., 2012; May et al., 2017). To understand the core similarity, we compared the protein sequences with well-characterized Merocyclophane cluster proteins from UIC10110 (Accession: AQA28562; May et al. 2017) and also compared with closely related *H. hongdechloris* species (Fig. 3B). The protein alignment shows that this gene has close homology to known Merocyclophane C domains (Supplementary Figure S4) and carried all major core domains associated with the PKS type-I Merocyclophane protein (Fig. 4B). However, *P. punensis* carries an extra domain involved in thioester reductase (TD), while *Nostoc* Sp. UIC10110 carries an extra methyltransferase domain. The presence of TD domains at the C terminal of the PKS gene cluster substantiates its role as a terminating enzyme and an important source of structural diversity in polyketide biosynthesis (Nakamura et al. 2015). This analysis suggests that the *P. punensis* genome encodes Type I polyketide synthases that are potentially responsible for Merocyclophane C and D; however, to functionally

validate these properties, further experiments will be required to verify its cellular function and antiproliferative activities.

Another putative protein involved in the biosynthesis of the 1-heptadecene pathway was identified (Position 18,422 – 24, 796). The 1-heptadecene is one of the common alkanes (hydrocarbons) produced by cyanobacteria (Gibbons et al. 2018) and is derived from octadecanoic acid (stearic acid; $C_{18}H_{36}O_2$). Heptadecene is considered an important constituent of hydrocarbons for biofuel production, and overexpression of intermediate pathway genes (*acyl-ACP*) in cyanobacteria (*Synechocystis* sp. PCC 6803) improves alkane production (Kudo et al. 2019). We compared the 1-heptadecene production in publicly available cyanobacteria species (Coates et al. 2014) with *P. punensis*, and this suggests that some species (*Cyanothece* sp. PCC 7822 and *Pleurocapsa* sp. PCC 7516) produce a high percentage of 1-heptadecene associated with the hydrocarbon products concentration (Coates et al. 2014). We identified that *P. punensis* could produce up to 31% of octadecanoic acid (precursor of heptadecene), suggesting a potential source of biofuel production. This observation provides an opportunity to study this gene and metabolic pathway in more detail (Table 3 and 4).

3.8. Multidrug and heavy metal related genes

Cyanobacteria can grow in diverse water conditions, including high salt concentrations, alkaline, heavy metal, and polluted water due to specific resistance genes that protect them against harmful chemicals from environmental sources (Singh 2014; Deshmukh et al. 2021). In particular, studying the multi-drug resistance proteins present in freshwater organisms is important as they pose a global concern for public health and the environment; however, these genes have not been investigated in great detail in fresh or marine water algae. We searched for multi-drug and heavy metal resistance genes in the *P. punensis* genome and identified 21 genes associated with multi-drug resistance (Supplementary Table S3). Among 21 genes, seven were associated with ABC transport and ATP-binding activity, while several others were associated with a different class of multi-drug resistance (MdtA, B, C, E, G, H, K, and N).

Cyanobacteria cells change their physiology and morphology after coming into contact with heavy metals and have been used to screen heavy metals in water sources (Samecka-Cymerman and Kempers, 2000; Kunasakaran et al., 2014). Recently, it has been identified that *Anabaena cylindrica* can tolerate higher concentrations of Cu, Pb, and Cd and produced the highest photosynthetic fluorescence when cultured for 7 days (Jusoh et al. 2020). We identified 12 genes associated with heavy metal tolerance (four arsenical, five cobalt-zinc, and three others) in *P. punensis* (Supplementary Table 2). To gain initial insights into the resilience of *P. punensis* to the most common heavy metals in freshwater (Mane et al. 2013), we conducted an experiment where the filaments of *P. punensis* were cultured in growth media containing different concentrations of Cu, Cd, Pb, and Zn (0, 1, 2, 2.5, and 5 mg/l). After 15 days of culture, we observed that all the samples tolerated the higher (5 mg/l) concentrations and were actively dividing. The presence of *P. punensis* might indicate heavy metal availability in freshwater or municipal water distribution systems. However, a detailed study addressing *P. punensis* heavy metal tolerance level is needed. The heavy metal content of Pavana river water is higher than the permissible limit, particularly for Cd, Ni, Pb, Hg, and Cu (Mane et al., 2013; Jamdade and Gawande, 2017). These observations suggest that *P. punensis* have a mechanism to tolerate and/or avoid heavy metals in water bodies, and therefore, can grow in these water conditions. But at the same time, previous studies and our data raise the concern about the safety of drinking water contaminated with this and other cyanobacteria. With proactive planning and active management, the municipal water distribution system could control the risk associated with cyanobacterial blooms and heavy metal content in finished drinking water (Yeager and Carpenter, 2019). The cyanobacterial growth can be controlled by reducing the inflow of nutrients that supports algal growth and also filtering heavy metals (Gao

Table 3
Identification of secondary metabolite biosynthesis gene clusters in *P. punensis* genome sequence.

Contigs	Location	Product type	Most similar known cluster	Similarity (%)
1	154,995–155,540	Bacteriocin	Pneumococcal choline-binding protein A	40
1	741,771–747,818	T1PKS	Merocyclophane C and D	66
3	18,422–24,796	T1PKS	1-Heptadecene	100
6	271,004–271,933	Terpene	Phytoene Synthesis	2
9	98,061–201,426	NRPS	Nonribosomal peptide synthetases (NRPS)	16

T1PKS: Type 1 polyketide synthase

NRPS: non-ribosomal peptide synthetase cluster

Table 4

Fatty acid profiling of *P. punensis*. The fatty acids with differential accumulation (with in row) are color coded. Red is lowest and green with highest concentration.

#	Compound name	BG11 media	BG11 biomass	PCMC water	PCMC biomass	Note
		Peak area (%)				
1	Octane (C ₈ H ₁₈)	2.34	1.48	2.03	1.34	Released in media
2	Nonane (C ₉ H ₂₀)	1.71	0.98	1.4	0.88	
3	Benzenethanamine, N,α,α trimethyl (C ₁₁ H ₁₇ N)	2.55	1.44	0.53	2.29	
4	Cyclooctasiloxane, hexadecamethyl (C ₁₆ H ₄₈ O ₈ Si ₈)	0.53	4.59	1.05	0.32	
5	Hexadecanoic acid, methyl ester (C ₁₇ H ₃₄ O ₂)	0.62	1.7	0	2.04	
6	cis-10-Heptadecenoic acid (C ₁₇ H ₃₂ O ₂)	1.37	1.77	1.58	1.88	
7	n-Hexadecanoic acid (C ₁₆ H ₃₂ O ₂)	7.82	9.58	8.24	11.13	
8	Dibutyl phthalate (C ₁₆ H ₂₂ O ₄)	1.21	1.72	2.05	0	
9	Nonadecatriene-5,14-diol (C ₁₉ H ₃₄ O ₂)	0	2.25	0	0	Specific to BG11 biomass
10	9-octadecen-4-olide (C ₁₈ H ₃₂ O ₂)	0	3.58	0	0	
11	Methyl 5,11,14-eicosatrienoate (C ₂₁ H ₃₆ O ₂)	0	1.37	0	0	
12	9-Octadecenoic acid, methyl ester, (C ₁₉ H ₃₆ O ₂)	0	3.02	0	1.84	Only in biomass
13	9,12-Octadecadienoic acid (C ₁₈ H ₃₂ O ₂)	12.35	12.88	13.02	18.9	
14	9-Octadecenoic acid (C ₁₈ H ₃₄ O ₂)	24.88	31.08	28.14	31.24	
15	Octadecanoic acid (C ₁₈ H ₃₆ O ₂)	4.19	5.91	5	5.67	
16	Cyclododecasiloxane eicosamethyl (C ₂₀ H ₆₀ O ₁₀ Si ₁₀)	3.38	1.42	2.31	0.77	Released in media
17	Glycidyl oleate (C ₁₉ H ₃₆ O ₃)	1.57	1.25	3.55	2.85	
18	Bis(2-ethylhexyl) phthalate (C ₂₄ H ₃₈ O ₄)	11.44	1.97	1.67	7.41	Released in BG11 media
19	Dibutyl phthalate (C ₁₆ H ₂₂ O ₄)	1.21	1.72	0	0	Specific to BG11 media
20	Tetracosamethyl-cyclododecasiloxane (C ₂₄ H ₇₂ O ₁₂ Si ₁₂)	6.67	1.53	6.87	0.98	Released in media

et al. 2018). Besides, assessing source water, monitoring the seasonal outbreak of algal blooms, making necessary treatment adjustments to filtered water and communicating with the public, can reduce the risks of cyanotoxins and heavy metals in drinking water (Yeager and

Carpenter, 2019).

The genes identified in this study might play a role in understanding heavy metal tolerance, uptake, or avoidance. However, these inferences await detailed study, including molecular, biochemical, and

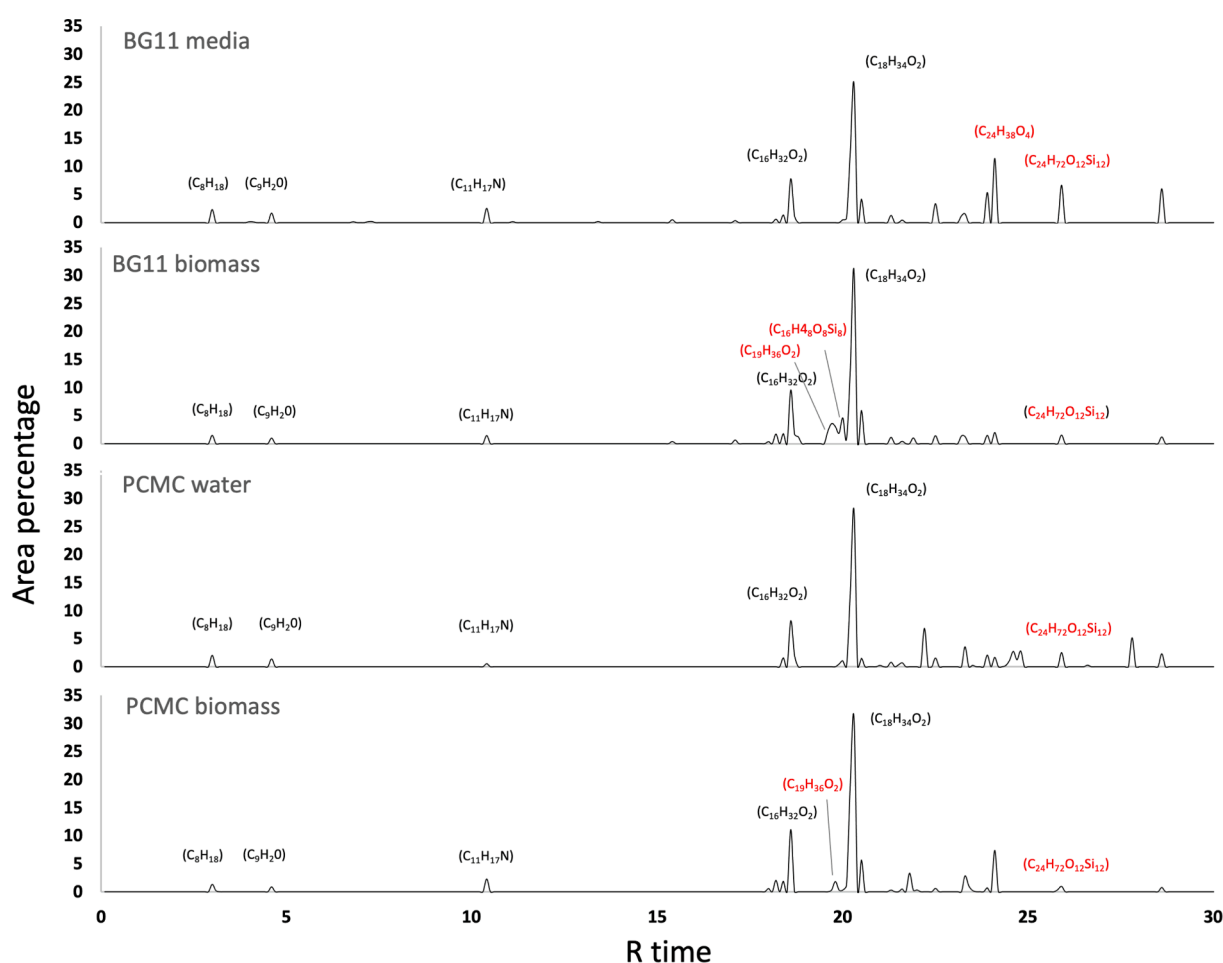


Fig. 5. GC–MS analysis of *P. punensis* fatty acids in BG11 media, BG11 biomass, PCMC water and PCMC biomass. Compounds with differential fatty acid composition are high-lighted in red text (for details see Table 4). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

physiological investigations, but the preliminary data and prospects are encouraging. Similarly, virulence factors and lipopolysaccharide (LPS) are cyanobacteria characteristics and provide resistance against other pathogenic bacteria and protect against abiotic and biotic stresses (Liu et al. 2019; Stewart et al. 2006). The virulence factors and LPS are referred to as potential toxins released in water and associated with various health effects in humans and animals (Stewart et al., 2006). Searches for virulence factor genes identified 57 genes, enriched for *Legionella pneumophila* (14%), followed by *Helicobacter pylori* (12%) and *Listeria monocytogenes* (12%) (Supplementary Figure S5). Likewise, based on gene ontology, we identified 12 genes associated with LPS activity in *P. punensis*, and these genes might have a direct or indirect role in exerting toxins into the water bodies (Supplementary Table S3). Nevertheless, we identified that *P. punensis* carries several potential genes that can reduce the quality of freshwater supplies. Therefore, *P. punensis* should be further studied.

3.9. Fatty acid content

The fatty acid composition of individual species of cyanobacteria is relatively conserved, and it is frequently utilized for industrial applications, such as biodiesel production, stimulants, and other purposes. The fatty acid composition also provides information on phylogenetic relationships of organisms (Kenyon et al., 1972; Los and Mironov, 2015; Shinde et al., 2016). The GC-MS analysis was performed to profile the fatty acid in the samples prepared from biomass grown in BG-11 medium and freshwater (Table 4). Additionally, the fatty acid content in the media (BG-11 and PCMC water) was estimated to study the release of fatty acids. A total of 40 different fatty acids were detected based on matches of the compound's mass spectra referred in the NIST library (<https://www.nist.gov/nist-research-library>) with great variation in the number of carbon atoms (C8-C24). Among these, 20 compounds were significant and detected in common or specifically in one or the other media compositions (Table 4, Fig. 5). For example, monosaturated 9-octadecenoic acid (18:1) was identified as a major fatty acid that showed 31.24% and 31.08% accumulation in the sample prepared from biomass grown in freshwater and BG-11 medium, respectively (Fig. 5). The 18:1 concentration in cyanobacteria species varies greatly within cyanobacteria species and is also affected by micro-environments and media composition (Sallal et al. 1990; Rajeshwari and Rajashekhar 2011).

Interestingly, we identified that the release of some of the compounds, like Bis (2-ethylhexyl) phthalate (#19 in Table 4), at relatively higher concentrations in BG-11 media (11.44%) compared to BG-11 biomass (1.97%). However, this compound was not released into PCMC water (1.67%) from its biomass (7.41%), and it could be because this colorless and viscous compound is soluble in oil and other media but not in water. This compound is considered an environmental toxicant and is widely used as a plasticizer, hydraulic fluid, and solvent in glowsticks, and cyanobacteria species, such as *Anabaena* sp. and *Oscillatoria* sp. are known to produce these compounds (Sarkar and Gourishankar, 2019). Tetracosamethyl-cyclododecasiloxane, Cyclo-decasiloxan eicosamethyl, Octane, and Nonane were specifically released in the external media (#20, 16, 1, 2, respectively in Table 4). Kaczmarzyk and Fulda (2010) also observed the release of a pool of unsaturated fatty acids into the culture medium from membrane lipids. Some compounds were identified specifically in BG11 or PCMC biomass, suggesting the effect of nutrient composition on the biosynthesis of these compounds (Table 4). Based on the correlation between morphological and physiological characteristics and cellular fatty acid composition, cyanobacteria strains were initially divided into four groups according to the content of the major fatty acid with the highest degree of unsaturation (Kenyon et al. 1972, Murata et al. 1992). In recent years, the validity of this classification has been supported and improved by genomic data (Los and Mironov 2015). Based on these previous classifications, *P. punensis* falls under Group 3 filamentous

cyanobacteria that contains a blend of saturated and unsaturated fatty acids. GO term enrichment for fatty acid biosynthesis, metabolism, and degradation identified 22 genes regulating 19 intermediate enzymes (Supplementary Figure S6). The concentration of most common fatty acids in *P. punensis* (average based on both biomasses) was palmitic acid (C₁₆H₃₂O₂; 10.35%), linoleic acid (C₁₈H₃₂O₂; 15.89%), Oleic acid (C₁₈H₃₄O₂; 31.16%), and Stearic acid (C₁₈H₃₆O₂; 5.79%) (Fig. 5). Although the information about fatty acid composition in other Pseudanabaenaceae is limited (Zimba et al. 2016), our analysis suggests that *P. punensis* content contains several important fatty acids that possess biological activities and can be potential sources in industrial applications.

4. Conclusion

Among diverse beneficial microorganisms, cyanobacteria can be used in several industrial applications, including nutrition, pharma, energy, and agriculture. However, some of these cyanobacteria (if not managed properly) can threaten environmental and public health. Here we have described a novel filamentous non-heterocystous *Pseudanabaena* sp. *punensis* isolated from the open ends of PVC pipelines, primarily used to deliver freshwater for domestic use. Using a polyphasic approach, we could confirm the novelty of the *P. punensis* and suggests its polyphyletic assemblage. Whole-genome comparative analysis has revealed interesting features and the presence of several important genes associated with the synthesis of bioactive metabolites, multi-drug and heavy metal resistance, and toxin-producing genes. This data suggests that *P. punensis* has several characteristics that can be potentially exploited in multiple industrial applications. However, their prevalence in freshwater water could impact environmental and public health. Therefore, further detailed functional and biochemical analyses should be done. Overall, the observations and inferences made in the present study will help limit its presence in the freshwater supply used for domestic purposes and provide a valuable genomic resource for future studies.

CRedit authorship contribution statement

Kirti M. Nitnaware: Formal morphological analysis, funding acquisition, investigation, writing original draft. **Kiran B. Raskar:** Methodology, field work, formal morphological analysis. **Gaurav Agarwal:** Data curation, analyzed whole genome sequence data. **Ricardo A. Chávez Montes:** analyzed whole genome sequence data. **Ratan Chopra:** analyzed whole genome sequence data. **Damar L. López-Arredondo:** Writing – reviewing and editing. **Tukaram D. Nikam:** Conceptualization, Supervision. **Gunvant B. Patil:** Conceptualization, funding acquisition, investigation, data interpretation, writing, reviewing and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics and consent to participate

This study did not involve humans, human data or animals; no ethics approval or consent is required to publish the results.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jympev.2021.107272>.

References

- Agasild, H., Panksep, K., Tönno, I., Blank, K., Kõiv, T., Freiberg, R., Laugaste, R., Jones, R.L., Nöges, P., Nöges, T., 2019. Role of potentially toxic cyanobacteria in crustacean zooplankton diet in a eutrophic lake. *Harmful Algae* 89, 101688. <https://doi.org/10.1016/j.hal.2019.101688>.
- Almagro Armenteros, J.J., Tsirigos, K.D., Sønderby, C.K., Petersen, T.N., Winther, O., Brunak, S., von Heijne, G., Nielsen, H., 2019. SignalP 5.0 improves signal peptide predictions using deep neural networks. *Nature Biotechnol.* 37 (4), 420–423. <https://doi.org/10.1038/s41587-019-0036-z>.
- Andreote, A.P.D., Vaz, M.G.M.V., Genuário, D.B., Barbiero, L., Rezende-Filho, A.T., Fiore, M.F., Raven, J., 2014. Nonheterocytous cyanobacteria from Brazilian saline-alkaline lakes. *J. Phycol.* 50 (4), 675–684. <https://doi.org/10.1111/jpy.12192>.
- Appels, R., Nystrom-Persson, J., Keeble-Gagnere, G., 2014. Advances in genome studies in plants and animals. *Funct. Integr. Genomics* 14 (1), 1–9. <https://doi.org/10.1007/s10142-014-0364-5>.
- Azevedo, M.T.V., Desouza, C.A., Rosado, T., Huszar, V., Roland, F., 2003. *Limnothrix bicudo*, a new species of Cyanophyceae/Cyanobacteria from Southeast of Brazil. *Algol. Stud.* 109 (4), 93–102. <https://doi.org/10.1127/1864-1318/2003/0109-0093>.
- Bankevich, A., Nurk, S., Antipov, D., Gurevich, A.A., Dvorkin, M., Kulikov, A.S., Lesin, V. M., Nikolenko, S.I., Pham, S., Prjibelski, A.D., Pyshkin, A.V., Sirotkin, A.V., Vyahhi, N., Tesler, G., Alekseyev, M.A., Pevzner, P.A., 2012. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J. Comput. Mol. Cell Biol.* 19 (5), 455–477. <https://doi.org/10.1089/cmb.2012.0021>.
- Basheva, D., Moten, D., Stoyanov, P., Belkinova, D., Mladenov, R., Teneva, I., 2018. Content of phycoerythrin, phycocyanin, allophycocyanin and phycoerythrocyanin in some cyanobacterial strains: Applications. *Eng. Life Sci.* 18 (11), 861–866.
- Bennett, A., Bogorad, L., 1973. Complementary chromatic adaptation in filamentous blue-green algae. *J. Cell Biol.* 58, 419–435. <https://doi.org/10.1083/jcb.58.2.419>.
- Blin, K., Wolf, T., Chevrete, M. G., Lu, X., Schwalen, C. J., Kautsar, S. A., Suarez Duran, H. G., De los Santos, E. L. C., Kim, H., Nave, M., Dickschat, J. S., Mitchell, D. A., Shelest, E., Breiting, R., Takano, E., Lee, S. Y., Weber, T., Medema, M. H., 2017. AntiSMASH 4.0—improvements in chemistry prediction and gene cluster boundary identification. *Nucl. Acids Res.* 45(W1), W36–W41. <https://doi.org/10.1093/nar/gkx319>.
- Boyer, S.L., Flechtner, V.R., Johansen, J.R., 2001. Is the 16S–23S rRNA internal transcribed spacer region a good tool for use in molecular systematics and population genetics? A case study in cyanobacteria. *Mol. Biol. Evol.* 18, 1057–1069. doi:10.1093/oxfordjournals.molbev.a003877.
- Bryant, D.A., Guglielmi, G., de Marsac, N.T., Castets, A.-M., Cohen-Bazire, G., 1979. The structure of cyanobacterial phycobilisomes: a model. *Arch. Microbiol.* 123 (2), 113–127. <https://doi.org/10.1007/BF00446810>.
- Caputo, A., Fournier, P.E., Raoult, D., 2019. Genome and pan-genome analysis to classify emerging bacteria. *Biol. Direct* 14(1), 5. <https://doi.org/10.1186/s13062-019-0234-0>.
- Castenholz, R. W., Rippka, R., Herdman, M., Wilmoth, A., 2015. Form- Pseudanabaena. *Bergey's Manual of Systematics of Archaea and Bacteria*, 1–5. doi:10.1002/9781118960608.gbm00444.
- Chang, A.C. G. Chen T., Li N., Duan J., 2019. Perspectives on Endosymbiosis in Coraloid Roots: Association of Cycads and Cyanobacteria. *Front. Microbiol.* 10, 1888 <https://doi.org/10.3389/fmicb.2019.01888>.
- Coates, R.C., Podell, S., Korobeynikov, A., Lapidus, A., Pevzner, P., Sherman, D.H., Allen, E.E., Gerwick, L., Gerwick, W.H., 2014. Characterization of cyanobacterial hydrocarbon composition and distribution of biosynthetic pathways. *PLOS one*, 9(1), p.e85140.
- Desikachary, T., 1959. *Cyanophyta*. ICAR Monographs on Algae, ICAR, New Delhi, p. 686.
- Deshmukh, R., Rana, N., Liu, Y., Zeng, S., Agarwal, G., Sonah, H., Varshney, R., Joshi, T., Patil, G.B., Nguyen, H.T., 2021. Soybean transporter database: A comprehensive database for identification and exploration of natural variants in soybean transporter genes. *Physiol. Plant.* 171 (4), 756–770.
- Elsaid, H., Naganuma, T., 2001. Phylogenetic diversity of ribulose-1, 5-bisphosphate carboxylase/oxygenase large-subunit genes from deep-sea microorganisms. *Appl. Environ. Microbiol.* 67 (4), 1751–1765. <https://doi.org/10.1128/AEM.67.4.1751-1765.2001>.
- Everroad, Craig, Wood, Michelle, 2006. Comparative molecular evolution of newly discovered picocyanobacterial strains reveals a phylogenetically informative variable region of β -phycoerythrin. *J. Phycol.* 42 (6), 1300–1311. <https://doi.org/10.1111/j.1529-8817.2006.00282.x>.
- Farrer, R.A., 2017. Synima: a Synteny imaging tool for annotated genome assemblies. *BMC Bioinf.* 18 (1), 507.
- Gao, J., Zhu, J., Wang, M., Dong, W., 2018. Dominance and growth factors of *Pseudanabaena* sp. in drinking water source reservoirs, Southern China. *Sustainability*, 10(11), p.3936.
- Genuário, D.B., De Souza, W.R., Monteiro, R.T.R., Sant'Anna, C.L., Melo, I.S., 2018. *Amazoninema* gen. nov. (Synecococcales, Pseudanabaenaceae) a novel cyanobacteria genus from Brazilian Amazonian rivers. *Int. J. Syst. Evolutionary Biol.* 68(7), 2249–2257. doi:10.1099/ijsem.0.002821.
- Gibbons, J., Gu, L., Zhu, H., Gibbons, W., Zhou, R., 2018. Identification of two genes required for heptadecane production in a N₂-fixing cyanobacterium *Anabaena* sp. strain PCC 7120. *AMB Express* 8 (1), 167.
- González-Morales, S.I., Pacheco-Gutiérrez, N.B., Ramírez-Rodríguez, C.A., Brito-Bello, A. A., Estrella-Hernández, P., Herrera-Estrella, L., López-Arredondo, D.L., 2020. Metabolic engineering of phosphite metabolism in *Synechococcus elongatus* PCC 7942 as an effective measure to control biological contaminants in outdoor raceway ponds. *Biotechnol. Biofuels* 13 (1), 1–19. <https://doi.org/10.1186/s13068-020-01759-z>.
- Gurevich, A., Saveliev, V., Vyahhi, N., Tesler, G., 2013. QUAST: Quality assessment tool for genome assemblies. *Bioinformatics* 29 (8), 1072–1075. <https://doi.org/10.1093/bioinformatics/btt086>.
- Harada, K.-I., 2004. Production of secondary metabolites by freshwater cyanobacteria. *Chem. Pharm. Bull.* 52 (8), 889–899. <https://doi.org/10.1248/cpb.52.889>.
- Howe, K.L., Contreras-Moreira, B., De Silva, N., Maslen, G., Akanni, W., Allen, J., Alvarez-Jarreta, J., Barba, M., Bolser, D.M., Cambell, L., Carbajo, M., 2020. Ensembl genomes 2020—enabling non-vertebrate genomic research. *Nucl. Acids Res.* 48 (D1), pp.D689–D695.
- Jamdade A. B., Gawande S. M., 2017. Analytical Study of Heavy Metal in Pavana River and its Effect on Aqua Culture. *Int. J. Sci. Res.* 4 (6), 572–575.
- Jones, P., Binns, D., Chang, H.Y., Fraser, M., Li, W., McAnulla, C., McWilliam, H., Maslen, J., Mitchell, A., Nuka, G., Pesseat, S., Quinn, A., Sangrador-Vegas, A., Scheremetjew, M., Yong, S., Lopez, R., Hunter, S., 2014. InterProScan 5: Genome-scale protein function classification. *Bioinformatics* 30 (9), 1236–1240. <https://doi.org/10.1093/bioinformatics/btu031>.
- Jusoh, N., Kin, C.M., Shing, W.L., Hock, O.G., Niu, V.B.W., 2020. Bioindication of Heavy Metals in Aquatic Environment using Photosynthetic Pigments in Cyanobacteria. *S. Afr. J. Chem. Eng.* 34, 78–81. <https://doi.org/10.1016/j.sajce.2020.05.011>.
- Kaczmarzyk, D., Fulda, M., 2010. Fatty acid activation in cyanobacteria mediated by acyl-acyl carrier protein synthetase enables fatty acid recycling. *Plant Physiol.* 152 (3), 1598–1610. <https://doi.org/10.1104/pp.109.148007>.
- Kang, H.S., Santarsiero, B.D., Kim, H., Kronic, A., Shen, Q., Swanson, S.M., Chai, H., Kinghorn, A.D., Orjala, J., 2012. Merocyclophanes A and B, antiproliferative cyclophanes from the cultured terrestrial Cyanobacteria *Nostoc* sp. *Phytochemistry* 79, 109–115. <https://doi.org/10.1016/j.phytochem.2012.03.005>.
- Kenyon, C.N., Rippka, R., Stanier, R.Y., 1972. Fatty acid composition and physiological properties of some filamentous blue-green algae. *Archiv. Mikrobiol.* 83 (3), 216–236. <https://doi.org/10.1007/BF00645123>.
- Khan, Z., Omar, W.M.W., Merican, F.M.M.S., Convey, P., Foong, C.P., Najimudin, N., 2018. Characterisation of *Pseudanabaena amphigranulata* (Synecococcales) isolated from a man-made pond, Malaysia: a polyphasic approach. *J. Appl. Phycol.* 30 (6), 3187–3196. <https://doi.org/10.1007/s10811-018-1392-7>.
- Kim, M., Chun, J., 2014. 16S rRNA gene-based identification of bacteria and archaea using the EzTaxon server. *Methods Microbiol.* 41, 61–74. Academic Press.
- Kitazaki, C., Numano, S., Takanezawa, A., Nishizawa, T., Shirai, M., Asayama, M., 2013. Characterization of lysis of the multicellular cyanobacteria *Limnothrix/Pseudanabaena* sp. strain ABRG5-3. *Biosci. Biotechnol. Biochem.* 77 (12), 2339–2347. <https://doi.org/10.1271/bbb.130409>.
- Komarek, J., Anagnostidis, K., 1986. Modern approach to the classification system of cyanophytes. 2-Chroococcales. *Algol. Stud.* 73, 157–226.
- Komárek, J., Anagnostidis, K., 2005. Cyanoprokaryota, Part 2: Oscillatoriales. In: Büdel, B., Krienitz, L., Gärtner, G., Schagerl, M. (Eds.), *Süßwasserflora von Mitteleuropa*. Elsevier Spektrum Akademischer Verlag, München, pp. 1–759.
- Kudo, H., Hayashi, Y., Arai, M., 2019. Improving hydrocarbon production by engineering cyanobacterial acyl-(acyl carrier protein) reductase. *Biotechnol. Biofuels* 12 (1), 291.
- Kumar, S., Stecher, G., Tamura, K., 2016. MEGA7: Molecular Evolutionary Genetics Analysis Version 7.0 for Bigger Datasets. *Mol. Biol. Evol.* 33 (7), 1870–1874. <https://doi.org/10.1093/molbev/msw054>.
- Kunasakaran, R., Vijayan, S., Shing, W.L., Kin, C.M., 2014. The effects of pH and cell density to the responses of immobilized cyanobacteria for copper detection. *J. Life Sci. Technol.* 2 (2), 78–81.
- Lagesen, K., Hallin, P., Rødland, E.A., Stærfeldt, H.H., Rognes, T., Ussery, D.W., Ussery, D.W., 2007. RNAMmer: consistent and rapid annotation of ribosomal RNA genes. *Nucleic Acids Res.* 35 (9), 3100–3108. <https://doi.org/10.1093/nar/gkm160>.
- Liu, B., Zheng, D.D., Jin, Q., Chen, L.H., Yang, J., 2019. VFDB 2019: a comparative pathogenomic platform with an interactive web interface. *Nucleic Acids Res.* 47 (D1), D687–D692. <https://doi.org/10.1093/nar/gky1080>.
- Los, D.A., Mironov, K.S., 2015. Modes of fatty acid desaturation in cyanobacteria: an update. *Life* 5(1), 554–567. <https://dx.doi.org/10.3390/life5010554>.
- Lurling, M., Waajen, G., Lisette, N., Domis, D., 2015. Evaluation of several end-of-pipe measures proposed to control cyanobacteria. *Aquat. Ecol.* 50, 499–519. <https://doi.org/10.1007/s10452-015-9563-y>.

- Maestro, B., Sanz, J.M., 2016. Choline binding proteins from *Streptococcus pneumoniae*: a dual role as enzybiotics and targets for the design of new antimicrobials. *Antibiotics* 5 (2), 21. <https://doi.org/10.3390/antibiotics5020021>.
- Mane, A.V., Pardeshi, R.G., Gore, V.R., Walave, R.L., Manjekar, S.S., Sutar, G.N., 2013. Water quality and sediment analysis at selected locations of Pavana river of Pune district, Maharashtra. *J. Chem. Pharm. Res.* 5 (8), 91–102.
- Martins, M. D., Branco, L. H., 2016. Potamolinea gen. nov. (Oscillatoriales, Cyanobacteria): a phylogenetically and ecologically coherent cyanobacterial genus. *Int. J. Syst. Evolut. Microbiol.* 66, 3632–3641. <https://doi.org/10.1099/ijsem.0.001243>.
- May, D.S., Chen, W.L., Lantvit, D.D., Zhang, X., Kronic, A., Burdette, J.E., Eustaquio, A., Orjala, J., 2017. Merocyclophanes C and D from the cultured freshwater cyanobacterium *Nostoc* sp. (UIC 10110). *J. Natural Products*, 80(4), 1073–1080. <https://doi.org/10.1021/acs.jnatprod.6b01175>.
- Murata, N., Wada, H., Gombos, Z., 1992. Modes of Fatty-Acid Desaturation in Cyanobacteria. *Plant Cell Physiol.* 33 (7), 933–941. <https://doi.org/10.1093/oxfordjournals.pcp.a078344>.
- Nakamura, H., Wang, J.X., Balskus, E.P., 2015. Assembly line termination in cylindrocyclophane biosynthesis: Discovery of an editing type II thioesterase domain in a type I polyketide synthase. *Chem. Sci.* 6 (7), 3816–3822.
- Newmaster, S.G., Fazekas, A.J., Ragupathy, S., 2006. DNA barcoding in land plants: evaluation of rbcL in a multigene tiered approach. *Botany* 84 (3), 335–341.
- Niiyama, Y., Tuji, A., Takemoto, K., Ichise, S., 2016. *Pseudanabaena foetida* sp. nov. and *P. subfoetida* sp. nov. (Cyanophyta/Cyanobacteria) producing 2-methylisoborneol from Japan. *Fottea* 16 (1), 1–11. <https://doi.org/10.5507/fof.2016.006>.
- Patel, R.K., Jain, M., Liu, Z., 2012. NGS QC toolkit: A toolkit for quality control of next generation sequencing data. *PLoS ONE* 7 (2), e30619. <https://doi.org/10.1371/journal.pone.0030619>.
- Patil, G.B., Lakhssassi, N., Wan, J., Song, L.i., Zhou, Z., Klepadlo, M., Vuong, T.D., Stec, A.O., Kahil, S.S., Colantonio, V., Valliyodan, B., Rice, J.H., Piya, S., Hewezi, T., Stupar, R.M., Meksem, K., Nguyen, H.T., 2019. Whole-genome re-sequencing reveals the impact of the interaction of copy number variants of the rhg1 and Rhg4 genes on broad-based resistance to soybean cyst nematode. *Plant Biotechnol. J.* 17 (8), 1595–1611. <https://doi.org/10.1111/pbi.2019.17.issue-810.1111/pbi.13086>.
- Rajeshwari, K.R., Rajashekhar, M., 2011. Biochemical composition of seven species of cyanobacteria isolated from different aquatic habitats of Western Ghats, Southern India. *Brazilian Arch. Biol. Technol.* 54 (5), 849–857. <https://doi.org/10.1590/S1516-89132011000500001>.
- Rippaka, R., Deruelles, J., Waterbury, J.B., 1979. Generic assignments, strain histories and properties of pure cultures of cyanobacteria. *J. General Microbiol.* 111, 1–61. <https://doi.org/10.1099/00221287-111-1-1>.
- Rodríguez-R, L. M., Gunturu, S., Harvey, W. T., Rosselló-Mora, R., Tiedje, J. M., Cole, J. R., Konstantinidis, K. T., 2018. The Microbial Genomes Atlas (MiGA) webserver: taxonomic and gene diversity analysis of Archaea and Bacteria at the whole genome level. *Nucl. Acids Res.* 46 (W1), W282–W288.
- Sallal, A.K., Nimer, N.A., Radwan, S.S., 1990. Lipid and fatty acid composition of freshwater cyanobacteria. *Microbiology* 136 (10), 2043–2048. <https://doi.org/10.1099/00221287-136-10-2043>.
- Samecka-Cymerman, A., Kempers, A. J., 2000. Bioindication of heavy metals with aquatic macrophytes: the case of a stream polluted with power plant sewages in Poland. *J. Toxicol. Environ. Health Part A* 31, 62 (1), 57–67. <https://doi.org/10.1080/00984100050201677>.
- Sarkar, A., Gourishankar, B.A., 2019. Computational optimization of in vitro parameters for di-(2-ethylhexyl) phthalate production from *Anabaena circinalis*. *J. Cell. Biochem.* 120 (1), 790–798. <https://doi.org/10.1002/jcb.v120.110.1002/jcb.27439>.
- Schattner, P., Brooks, A.N., Lowe, T.M., 2005. The tRNAscan-SE, snoscan and snoGPS web servers for the detection of tRNAs and snoRNAs. *Nucleic Acids Res.* 33 (SUPPL. 2), 686–689. <https://doi.org/10.1093/nar/gki366>.
- Seemann, T., 2014. Prokka: Rapid prokaryotic genome annotation. *Bioinformatics* 30 (14), 2068–2069. <https://doi.org/10.1093/bioinformatics/btu153>.
- Shih, P.M., Wu, D., Latifi, A., Axen, S.D., Fewer, D.P., Talla, E., Calteau, A., Cai, F., De Marsac, N.T., Rippaka, R., Herdman, M., 2013. Improving the coverage of the cyanobacteria phylum using diversity-driven genome sequencing. *Proc. Natl. Acad. Sci.* 110 (3), 1053–1058. <https://doi.org/10.1073/pnas.1217107110>.
- Singh, D., Chandrababunaidu, M.M., Panda, A., Sen, D., Bhattacharyya, S., Adhikary, S. P., Tripathy, S., 2015. Draft genome sequence of cyanobacterium *Hassallia byssoides* strain VB512170, isolated from Monuments in India. *Genome Announcements* 3 (2). <https://doi.org/10.1128/genomeA.00064-15>.
- Simão, F.A., Waterhouse, R.M., Ioannidis, P., Kriventseva, E.V., Zdobnov, E.M., 2015. BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics* 31 (19), 3210–3212.
- Shinde, S., Villamor, J.G., Lin, W., Sharma, S., Verslues, P.E., 2016. Proline coordination with fatty acid synthesis and redox metabolism of chloroplast and mitochondria. *Plant Physiol.* 172 (2), 1074–1088. <https://doi.org/10.1104/pp.16.01097>.
- Singh, S., 2014. A review on possible elicitor molecules of cyanobacteria: their role in improving plant growth and providing tolerance against biotic or abiotic stress. *J. Appl. Microbiol.* 117 (5), 1221–1244. <https://doi.org/10.1111/jam.12612>.
- Stewart, I., Schluter, P. J., Shaw, G.R., 2006. Cyanobacterial lipopolysaccharides and human health—a review. *Environ. Health.* 5(1), 1–23. <https://dx.doi.org/10.1186%2F1476-069X-5-7>.
- Tamura, K., Peterson, D., Peterson, N., Stecher, G., Nei, M., Kumar, S., 2011. MEGA5: Molecular Evolutionary Genetics Analysis Using Maximum Likelihood, Evolutionary Distance, and Maximum Parsimony Methods. *Mol. Biol. Evol.* 28 (10), 2731–2739. <https://doi.org/10.1093/molbev/msr121>.
- Taylor, E.L., Taylor, T.N., Krings, M., 2009. *Paleobotany: the biology and evolution of fossil plants*. Academic Press.
- Tuji, A., Niiyama, Y., 2018. Two new *Pseudanabaena* (Cyanobacteria, Synechococcales) species from Japan, *Pseudanabaena cinerea* and *Pseudanabaena yagii*, which produce 2-methylisoborneol. *Phycol. Res.* 66, 291–299. <https://doi.org/10.1111/pre.12327>.
- Turland, N. J., Wiersma, J. H., Barrie, F. R., Greuter, W., Hawksworth, D. L., Herendeen, P. S., Knap, S., Kusber, W.-H., Li, D.-Z., Marhold, K., May, T. W., McNeill, J., Monro, A. M., Prado, J., Price, M. J., Smith, G. F. (Eds.) 2018: International Code of Nomenclature for algae, fungi, and plants (Shenzhen Code) adopted by the Nineteenth International Botanical Congress Shenzhen, China, July 2017. *Regnum Vegetabile* 159. Glashütten: Koeltz Botanical Books. DOI <https://doi.org/10.12705/Code.2018>.
- Valliyodan, B., Qiu, D., Patil, G., Zeng, P., Huang, J., Dai, L., Chen, C., Li, Y., Joshi, T., Song, L., Vuong, T.D., 2016. Landscape of genomic diversity and trait discovery in soybean. *Sci. Rep.* 6 (1) <https://doi.org/10.1038/srep23598>.
- Vincent, R.K., Qin, X., McKay, R.M.L., Miner, J., Czajkowski, K., Savino, J., Bridgeman, T., 2004. Phycocyanin detection from LANDSAT TM data for mapping cyanobacterial blooms in Lake Erie. *Remote Sens. Environ.* 89 (3), 381–392. <https://doi.org/10.1016/j.rse.2003.10.014>.
- Walker, D., Fathabad, S. G., Tabatabai, B., Jafar, S., Sittler, V., 2019. Microcystin Levels in Selected Cyanobacteria Exposed to Varying Salinity. *J. Water Resource Protect.*, 11, 395–403. <https://dx.doi.org/10.4236/2Fjwarp.2019.114023>.
- Whitton, B.A., Potts, M. (Eds.), 2000. *The ecology of cyanobacteria: their diversity in time and space*. Kluwer Academic Publishers, Dordrecht, the Netherlands, ISBN 0792 347358.
- Yeager, N., Carpenter, A., 2019. State approaches to addressing cyanotoxins in drinking water. *AWWA Water Science*, 1(1), p.e1121.
- Yoon, S.H., Ha, S.M., Kwon, S., Lim, J., Kim, Y., Seo, H., Chun, J., 2017. Introducing EzBioCloud: A taxonomically united database of 16S rRNA gene sequences and whole-genome assemblies. *Int. J. Syst. Evol. Microbiol.* 67 (5), 1613–1617. <https://doi.org/10.1099/ijsem.0.001755>.
- Zehr, J.P., 2011. Nitrogen fixation by marine cyanobacteria. *Trends Microbiol.* 19 (4), 162–173. <https://doi.org/10.1016/j.tim.2010.12.004>.
- Zimba, P. V., Huang, I. S., Foley, J. E., Linton, E. W., 2016. Identification of a new-to-science cyanobacterium, *Toxifilum mysidocida* gen. nov. & sp. nov. (Cyanobacteria, Cyanophyceae) *J. Phycol.* 53(1), 188–197. doi: 10.1111/jpy.12490.
- Zhu, M., Zhou, P.P., Yu, L.J., 2002. Extraction of lipids from *Mortierella alpina* and enrichment of arachidonic acid from the fungal lipids. *Bioresour. Technol.* 84, 93–95.